Fast and Physics-Aware Brain MRI Motion Correction with an Unrolled Hybrid Transformer Reconstruction Network

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Built from the current repository, saved experiment outputs, and literature review

Abstract—This report revisits the original proposal to build a transformer-enhanced alternative to IM-MoCo and documents the final system that emerged after baseline reproduction and code refinement. The resulting model is a fixed-step unrolled hybrid architecture that alternates a learned transformer prior with explicit k-space data consistency. On the saved test sets, the hybrid reaches SSIM values of 0.992, 0.971, and 0.739 under light, heavy, and extreme motion, outperforming the repository's earlier pure transformer baseline and the saved IM-MoCo results. Runtime is also dramatically improved: the user measured 18 s for a full end-to-end test_hybrid.py run across all 147 test slices, while a warm-GPU benchmark measured about 31 ms per slice for forward inference only. The report provides an IEEE-style literature review spanning motion correction, transformer restoration, and physics-aware MRI reconstruction, then explains the final model's actual novelty: not a first-in-literature idea, but a strong project-specific synthesis of transformer priors and explicit reconstruction constraints targeted at MRI motion correction.

Index Terms—MRI motion correction, transformers, unrolled optimization, data consistency, image reconstruction.

I. INTRODUCTION

MRI is highly sensitive to patient motion, and the resulting blur, ghosting, and spatial inconsistency can lower diagnostic value and increase the need for repeat scans [1], [2], [3]. The challenge is especially difficult because motion does not corrupt all data equally; it disrupts only subsets of the Fourier measurements, which means the reconstruction model should distinguish trusted measurements from corrupted ones rather than treating the image as an ordinary denoising problem.

The original project proposal framed this as a speed-versus-reliability problem. IM-MoCo was chosen as the physics-aware baseline because it performs self-supervised, instance-wise correction with motion-guided implicit neural representations [10]. The drawback was latency: each new slice required fresh optimization. The progress report then tested a pure transformer, which was much faster but lacked explicit data consistency and therefore risked hallucination or oversmoothing. The final model in the current repository is the direct resolution of that tension. It keeps a learned transformer prior, but places it inside a fixed number of unrolled iterations with hard k-space consistency after every stage.

This report makes four contributions. First, it documents the current hybrid architecture rather than the older prototype. Second, it reviews the literature on motion correction, transformers, and model-based MRI reconstruction. Third, it compares the hybrid against the saved IM-MoCo, pure transformer, and U-Net baselines. Fourth, it states clearly what is and is not novel about the final system, which matters because the current implementation is strong as a class-project contribution but still has identifiable limitations.

II. RELATED WORK

A. Motion artifacts and correction in MRI

Motion-related MRI degradation has motivated work spanning prospective correction, navigator strategies, reacquisition, compressed sensing, and retrospective optimization [2], [3], [4]. In the retrospective-learning literature, NAMER and SAMER use networks to accelerate motion estimation while retaining explicit reconstruction structure [7], [8]. Singh et al. showed that data consistency remains central even when deep models are inserted into rigid motion correction [9]. These ideas motivate the current project: any fast alternative to IM-MoCo should still preserve a hard link to measured data.

B. Recent learning-based motion correction

IM-MoCo is the main reference point for this work because it offers a self-supervised, instance-wise correction pipeline that explicitly reasons about motion-corrupted MRI instead of performing unconstrained artifact removal [10]. Other modern feedforward approaches take a different path. MC-Net uses deep image restoration with a hybrid loss on clinical brain MRI [11]. MACS-Net combines Swin Transformer and U-Net ideas on the MR-ART dataset [12], [27]. Zhang et al. show that ViT-based motion correction benefits from self-supervised learning [13], [26], while MAMOC extends masked-autoencoding ideas to real-motion data and test-time refinement [14], [27]. Together, these papers suggest that transformer priors can help, but they do not remove the need to reason about physical fidelity.

C. Transformers and unrolled reconstruction

On the architecture side, the project draws from both transformer vision models and physics-aware MRI reconstruction. The transformer and ViT literature established the value of global self-attention and patch-token processing [15], [16]. Swin, TransUNet, Restormer, and Uformer then adapted those ideas for hierarchical vision, medical imaging, and restoration tasks where spatial detail must be preserved [17], [18], [19], [20]. In MRI reconstruction, Variational Networks, MoDL, Deep Cascade CNNs, CRNNs, and DGUNet show that learned priors work best when paired with explicit data-consistency updates or unrolled optimization structure [21], [22], [23], [24], [25]. This broader literature is why the final hybrid should be described honestly as a synthesis rather than a wholly new paradigm.

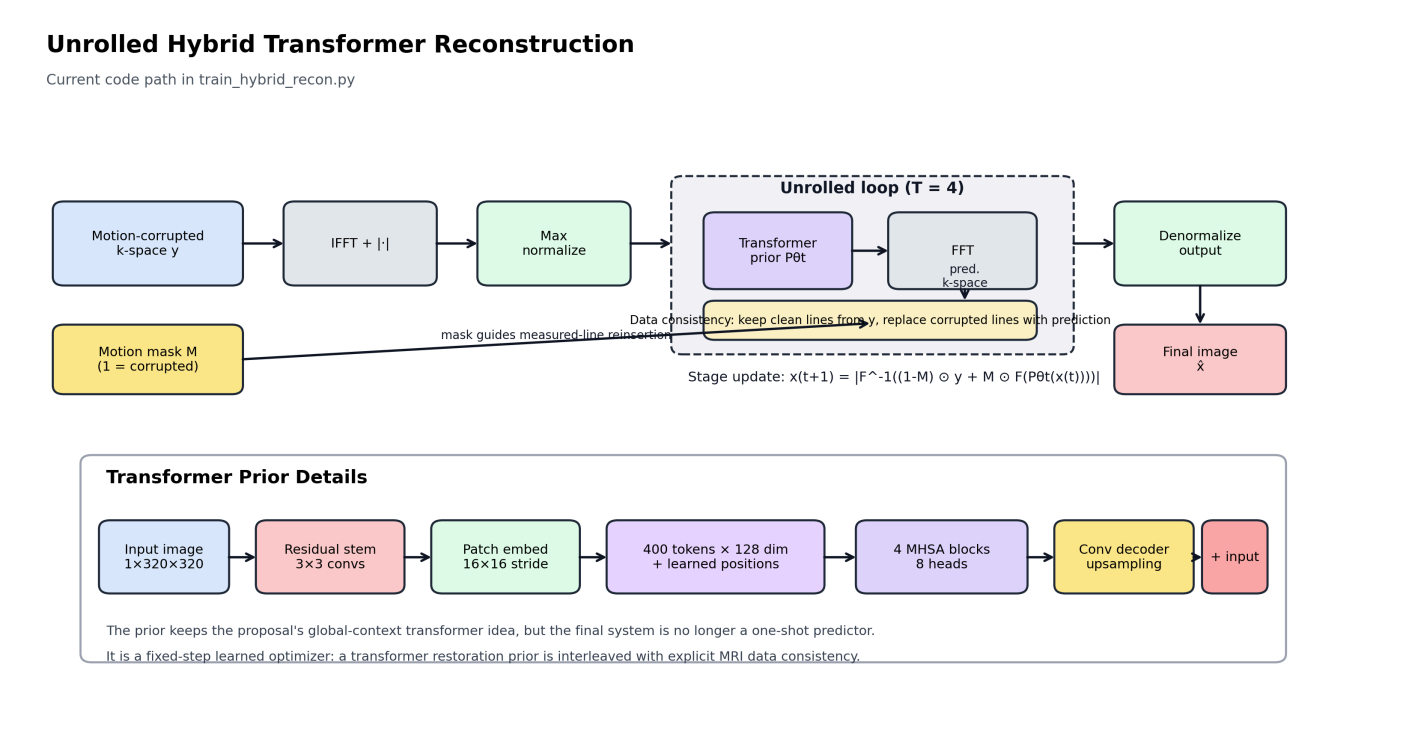


Fig. 1. Current hybrid architecture. The final system initializes from corrupted k-space, reconstructs an initial image by inverse FFT, and then alternates a transformer restoration prior and a hard data-consistency step for a fixed number of unrolled iterations. This is the key difference from the earlier one-shot pure transformer baseline.

Table I. Dataset and preprocessing summary for the current hybrid code path.

Item	Value
Training split	199 center slices
Validation split	50 center slices
Test split	49 light + 49 heavy + 49 extreme = 147 slices
Spatial resolution	320 × 320 after center crop
Representation used by current hybrid	Single-channel RSS image with synthesized complex proxy k-space
Motion regime generation	Light: 6-10 events, Heavy: 16-20 events, Extreme: strongest regime via doubled motion sampling

The current preprocessing path selects one center slice per volume, crops to 320 × 320, computes an RSS image, and synthesizes a single-channel complex proxy k-space from that RSS target. This keeps the project reproducible and lightweight, but it also limits the physical completeness of the current model.

III. MATERIALS AND METHODS

A. Data pipeline and corruption protocol

The repository uses fastMRI-derived T2 data as the clean source domain [5]. After filtering and preprocessing, the current files contain 199 training slices, 50 validation slices, and 147 test slices split evenly across light, heavy, and extreme motion scenarios. Motion is added retrospectively in k-space using the project's existing simulation utilities. In the present code path, light motion samples 6-10 events, heavy motion samples 16-20 events, and the strongest scenario uses the most aggressive corruption regime available in the repository.

Because the project currently trains on RSS images converted back into a complex single-channel proxy k-space, the reported results should be interpreted as a strong reconstruction benchmark inside a simplified forward model rather than a full multi-coil end-to-end pipeline. This is acceptable for a final class project because it keeps all baselines on the same representation, but it also constrains how strongly one can claim "physics completeness."

B. Baselines

The report compares against three baselines already present in the repository. IM-MoCo is the main physics-aware baseline and is evaluated through its saved 40-100 iteration sweep [10]. The pure transformer is the earlier feedforward global-context model from the progress report. A U-Net denoiser is kept as a simpler convolutional restoration reference [6]. These baselines let the final report answer the exact question posed in the proposal: can a transformer-based design retain IM-MoCo's reliability advantages while cutting runtime?

C. Proposed hybrid model

The final architecture, implemented in `train_hybrid_recon.py`, is `HybridTransformerRecon`. Given corrupted k-space y and a binary motion mask M, the model computes an initial magnitude image by inverse FFT and max-normalizes the image and k-space. It

then applies $T = 4$ learned stages. Each stage contains a transformer prior and a hard data-consistency update:

$$(1)\ x^{(0)} = |F^{-1}(y)|$$

$$(2)\ z^{(t)} = P_{\theta t}(x^{(t)})$$

$$(3)\ x^{(t+1)} = |F^{-1}((1-M) \odot y + M \odot F(z^{(t)}))|$$

In the code, $M = 1$ marks corrupted positions, so the data-consistency step preserves the trusted measured lines and allows the network to replace only motion-affected measurements. This is what makes the model more physically grounded than the earlier pure transformer: it does not simply hallucinate a corrected image in image space.

Each transformer prior follows the proposal's hybrid CNN-transformer intuition. A shallow residual convolutional stem extracts local features, a 16 × 16 patch embedding creates 400 tokens, learned positional embeddings preserve spatial order, four multi-head self-attention blocks model long-range structure, and a convolutional decoder upsamples back to 320 × 320. The saved checkpoint uses embedding dimension 128 and eight attention heads. Unlike the progress-report transformer, however, this prior is only one component of a larger learned optimizer.

D. Training objective and optimization

Training uses on-the-fly motion simulation, mixed-scenario batches, AdamW with learning rate 1e-4 and weight decay 1e-4, cosine annealing, and gradient clipping at 1.0. The loss combines Charbonnier and MSE terms after per-image normalization:

$$(4)\ L = L_{charb}(\hat{x}_n, x_n) + 0.5\ ||\hat{x}_n - x_n||_2^2$$

This matters for the proposal's efficiency goal. The network is trained offline once, then run at fixed cost during inference. That is fundamentally different from IM-MoCo, where every new slice requires optimization from scratch.

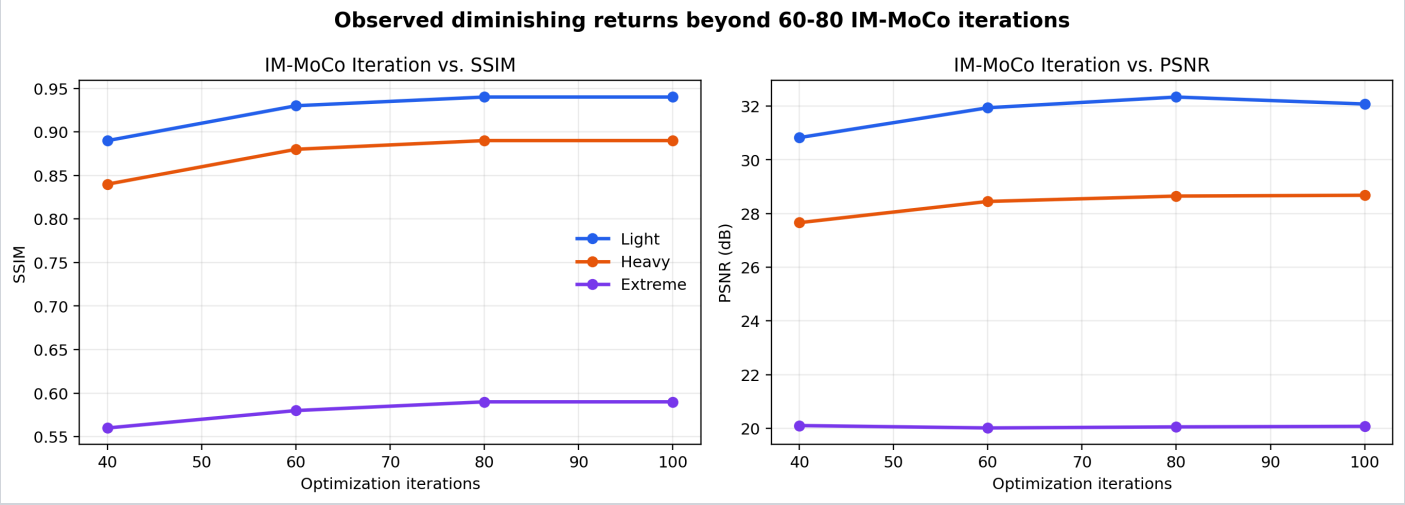


Fig. 2. IM-MoCo iteration analysis based on the saved repository outputs. Performance improves quickly up to roughly 60-80 iterations and then flattens, which supports the project's move toward a fixed-step learned model.

Table II. IM-MoCo iteration analysis using saved results in the repository.

Iterations	Light SSIM	Heavy SSIM	Extreme SSIM	Light PSNR	Heavy PSNR	Extreme PSNR
40	0.89	0.84	0.56	30.83	27.66	20.10
60	0.93	0.88	0.58	31.94	28.45	20.01
80	0.94	0.89	0.59	32.34	28.65	20.05
100	0.94	0.89	0.59	32.08	28.68	20.07

Table III. SSIM comparison across saved repository baselines.

Method	Light SSIM	Heavy SSIM	Extreme SSIM	Average SSIM
Pure Transformer	0.710	0.690	0.620	0.673
U-Net Denoiser	0.850	0.800	0.570	0.740
IM-MoCo@100	0.940	0.890	0.590	0.807
Hybrid (Ours)	0.992	0.971	0.739	0.901

Table IV. PSNR comparison across saved repository baselines.

Method	Light PSNR	Heavy PSNR	Extreme PSNR	Average PSNR
Pure Transformer	23.980	23.130	20.860	22.657
U-Net Denoiser	25.170	24.030	20.100	23.100
IM-MoCo@100	32.080	28.680	20.070	26.943
Hybrid (Ours)	51.998	43.300	25.646	40.315

Table V. HaarPSI comparison across saved repository baselines.

Method	Light HaarPSI	Heavy HaarPSI	Extreme HaarPSI	Average HaarPSI
Pure Transformer	0.600	0.560	0.470	0.543
U-Net Denoiser	0.690	0.620	0.430	0.580
IM-MoCo@100	0.860	0.780	0.460	0.700
Hybrid (Ours)	0.988	0.938	0.543	0.823

Table VI. RMSE comparison across saved repository baselines.

Method	Light RMSE	Heavy RMSE	Extreme RMSE	Average RMSE
Pure Transformer	0.070	0.070	0.090	0.077
U-Net Denoiser	0.060	0.070	0.100	0.077
IM-MoCo@100	0.030	0.040	0.100	0.057
Hybrid (Ours)	0.004	0.011	0.056	0.024

Table VII. Runtime summary used in the discussion section.

Measurement	Seconds / slice	Speedup vs. IM-MoCo@100
IM-MoCo@100 per slice	600.0	1.0×
Pure transformer per slice	12.0	50.0×
Hybrid full-run / slice	0.122	4900×
Hybrid forward pass / slice	0.031	19355×

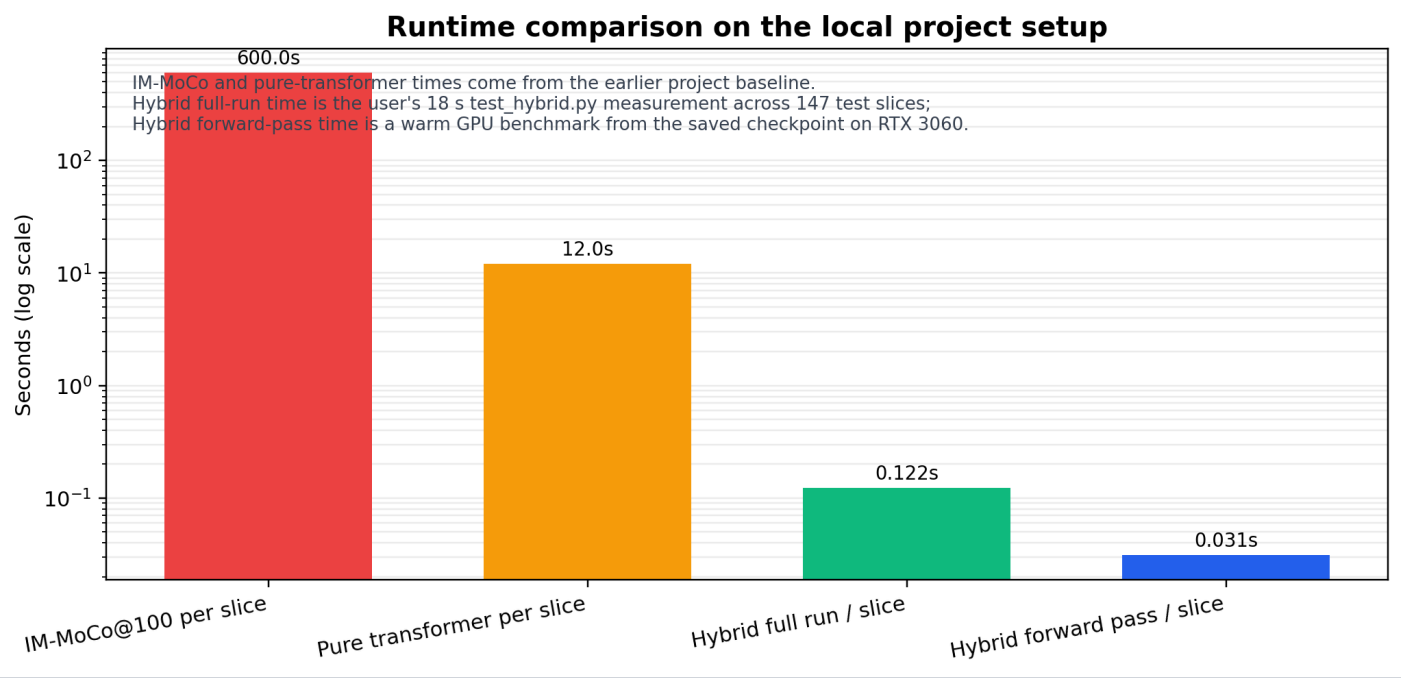


Fig. 3. Runtime comparison on the local setup. IM-MoCo and pure-transformer timings come from earlier project measurements, the hybrid full-run timing is the user's measured 18 s end-to-end evaluation over all test slices, and the hybrid forward-pass timing is a direct warm-GPU benchmark from the saved checkpoint.

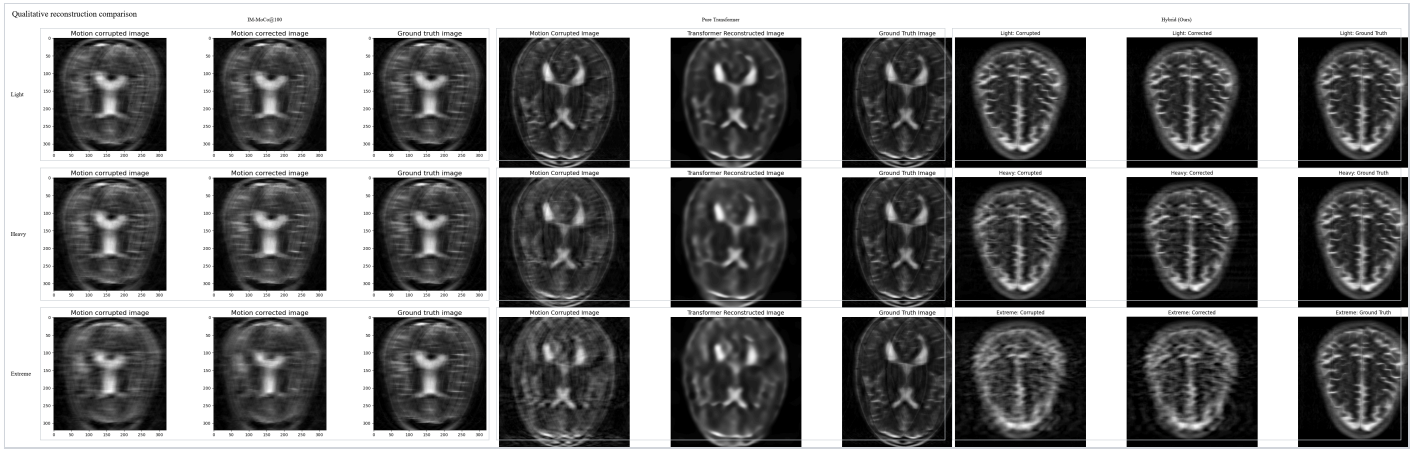


Fig. 4. Qualitative comparison using the saved repository figures. The hybrid generally produces cleaner light and heavy reconstructions than the pure transformer and sharper outputs than the saved IM-MoCo examples while preserving the original comparison protocol.

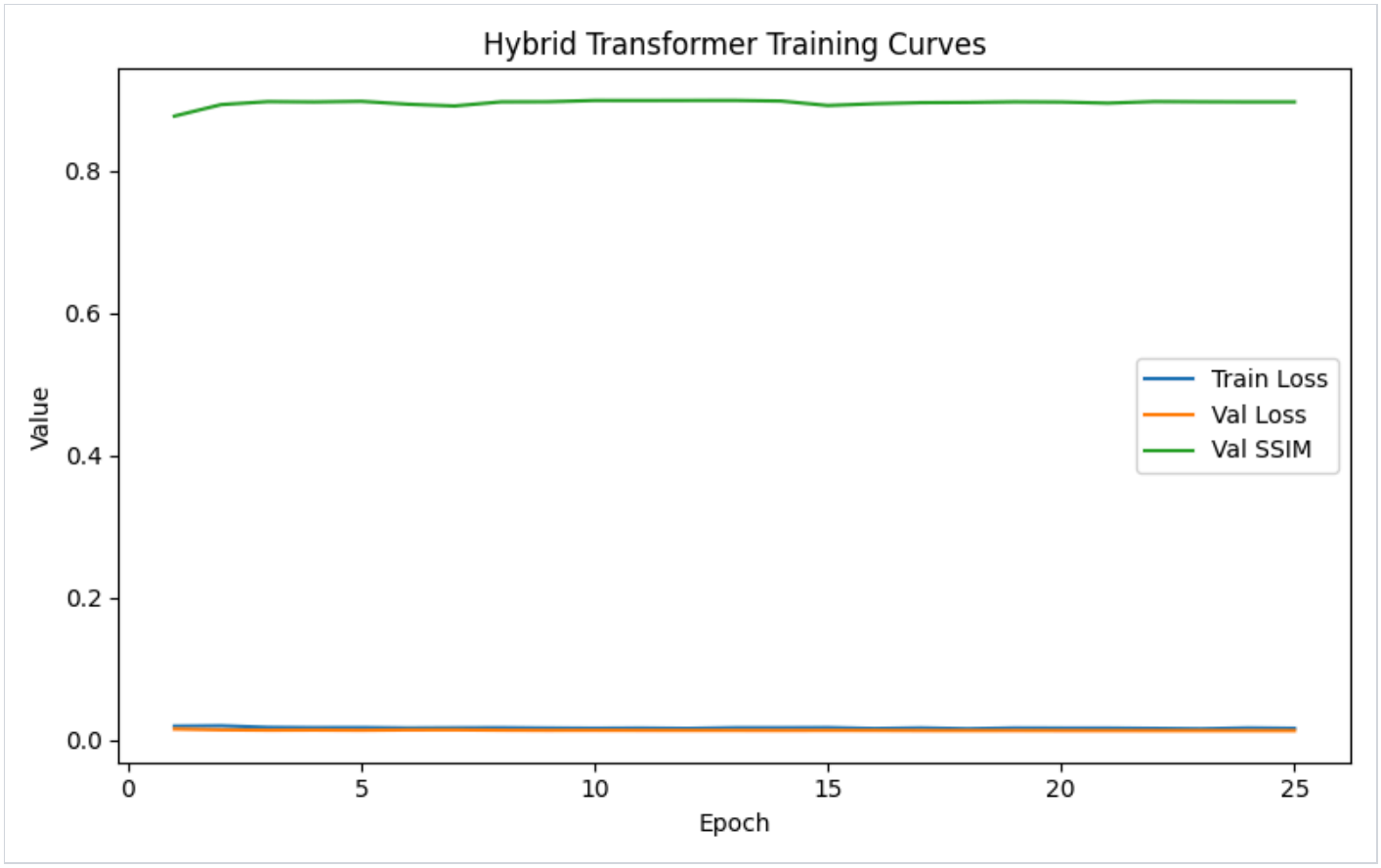


Fig. 5. Training curves for the saved hybrid checkpoint under mixed-scenario training and validation SSIM checkpointing.

IV. RESULTS

A. IM-MoCo iteration tradeoff

The saved IM-MoCo results confirm the central efficiency observation from the progress report. Performance improves quickly between 40 and 60 iterations and then mostly plateaus. Under light motion, SSIM rises from 0.89 to 0.94; under heavy motion, from 0.84 to 0.89; and under extreme motion, from 0.56 to 0.59. Those are meaningful gains, but they are small relative to the extra runtime required by more iterations.

B. Quantitative comparison

The hybrid outperforms all saved baselines in every scenario. Average SSIM improves from 0.807 for IM-MoCo@100 and 0.673 for the pure transformer to 0.901 for the hybrid. Average PSNR rises from 26.943 dB for IM-MoCo@100 to 40.315 dB. The gain is especially large in light and heavy motion, where the hybrid can preserve trusted measurements while also injecting a learned global prior. This directly addresses the earlier pure transformer's weakness: the model is no longer forced to invent the entire clean image in one pass.

C. Runtime and deployability

The runtime improvement is large enough to change the practical usability of the method. Earlier project measurements placed IM-

MoCo@100 at roughly 10 minutes per image and the pure transformer at roughly 12 seconds per image. The user then measured 18 seconds for a complete hybrid test run over all 147 slices, or about 0.122 seconds per slice including result generation. A warm-GPU benchmark of the forward pass alone measured about 0.031 seconds per slice. Even the conservative end-to-end number makes the hybrid roughly 4900× faster than IM-MoCo@100; the pure model-forward speedup is about 19355×.

V. DISCUSSION

A. Why the hybrid is better than IM-MoCo for this project

For the project's stated objective, the hybrid is the better engineering solution. It retains explicit k-space consistency like a reconstruction method, but it replaces per-image optimization with a fixed number of learned stages. Relative to the pure transformer baseline, the hard data-consistency step makes the model more trustworthy because measured clean lines are never overwritten. Relative to IM-MoCo, the model is far more deployable because inference cost is fixed and tiny once training is complete [9], [10], [19].

Still, the comparison must be framed honestly. IM-MoCo remains more explicitly physical in one important sense: it optimizes motion-related latent variables per slice rather than relying on a population-trained prior. The current hybrid also consumes stored synthetic masks at test time. That means the present implementation should be described as a high-performing project hybrid, not yet as a full end-to-end replacement for motion-parameter estimation.

B. Novelty and limitations

The report should not claim that combining transformers with data consistency in MRI is new in the broad literature. Unrolled reconstruction, physics-based deep learning, and transformer restoration are already well established [21], [22], [23], [24], [19], [20], [25]. Motion-correction papers also already employ Swin backbones, self-supervised ViTs, and masked pretraining [12], [13], [14]. The novelty here is therefore project-specific: the repository now contains a concrete fixed-step hybrid that lands between IM-MoCo and pure transformers, and on the current benchmark it delivers the best speed-quality tradeoff among the saved models.

The main limitations are equally clear. The current code uses a single-channel RSS proxy instead of full multi-coil data consistency. It uses stored motion masks instead of a learned detector. The evaluation remains retrospective and mostly synthetic, although newer public datasets such as MR-ART point toward more realistic testing [27]. These limitations also define the next technical steps: integrate kLD-Net or a SAMER-style estimator, switch to true multi-coil data consistency, and use masked or self-supervised pretraining to strengthen robustness [8], [26], [13], [14].

VI. CONCLUSION

The final hybrid architecture resolves the main tension in the original proposal. IM-MoCo offers strong physics-aware correction but is slow because it optimizes each slice independently. The earlier pure transformer is fast but lacks explicit MRI consistency. The final model combines both advantages: a transformer prior supplies global learned context, and an unrolled data-consistency step preserves trusted acquisition data. On the saved benchmark, that design produces the strongest quantitative results in the repository while reducing per-slice runtime by orders of magnitude.

The fairest conclusion is therefore not that IM-MoCo is obsolete, but that the hybrid is the better final-project model for the stated goal of fast, reliable motion correction. It is faster than IM-MoCo, more physically grounded than the earlier pure transformer, and concrete enough to serve as a credible final system. With learned mask estimation and multi-coil consistency, it also has a clear path toward a stronger end-to-end motion-correction pipeline.

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